

How important is
early life experience?



T/Th 9:35-10:25am

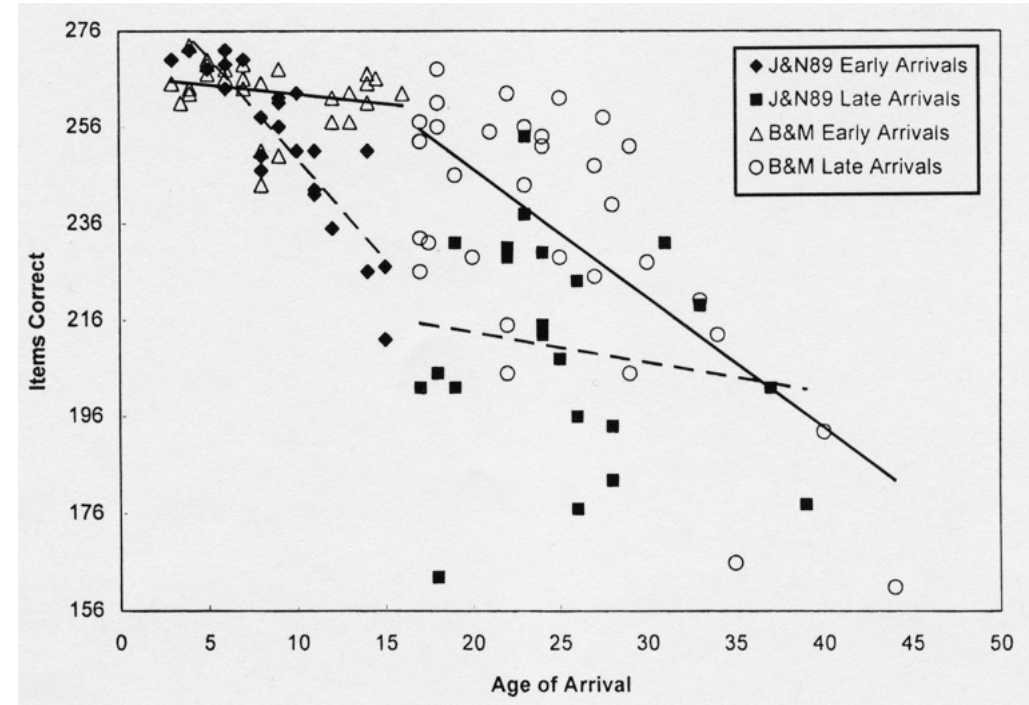
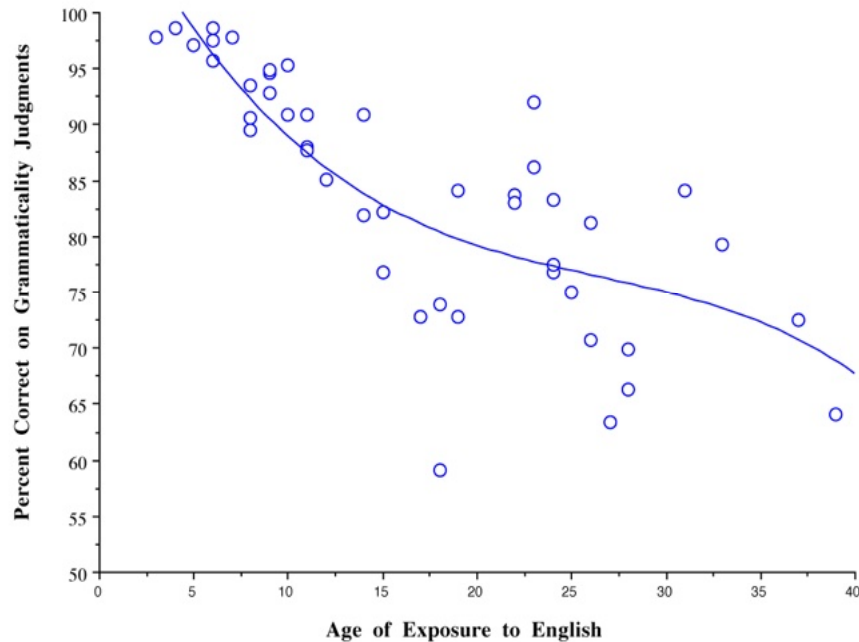
Professor Casey Lew-Williams

Kennedy Casey, Abby Fergus, Sarah Joo, Chantal Valdivia, Claire Whiting

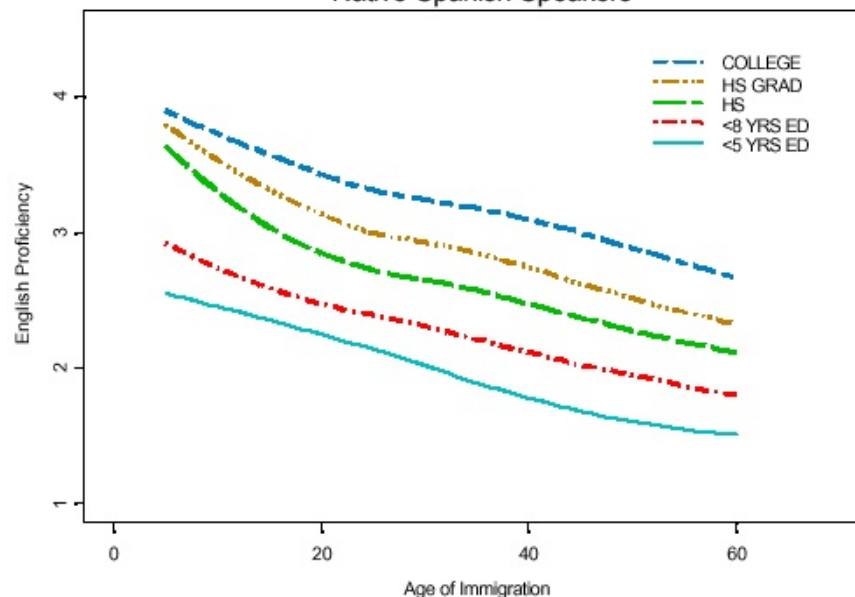
A moment on bilingualism & bidialectism

Is there a critical period for language development?

Johnson & Newport: 2nd Language Learning as a Function of Age of Exposure to English



English Proficiency by Age of Immigration
Native Spanish Speakers



(Birdsong & Molis, 2001; Hakuta & Bialystok, 1994; Hakuta et al., 2003; Johnson & Newport, 1989)

Common questions about early bilingualism

1. Are bilingual children confused?
2. Does bilingualism make children smarter?
3. Is it best for each parent to speak only one language with a bilingual child?
4. Should parents avoid mixing languages together?
5. Is earlier better?
6. Are bilingual children more likely to have language difficulties, delays, or disorders?

*For quick, evidence-based answers: Byers-Heinlein & Lew-Williams (2013).
Bilingualism in the early years: What the Science Says.*

Dialect: African American English

EDUCATION

Julie Washington's Quest to Get Schools to Respect African-American English

The speech pathologist believes that helping kids switch seamlessly between dialects is a key to their success.

WILLIAM BRENNAN APRIL 2018 ISSUE



Dialect: Black ASL



(Lucas, Bayley, McCaskill, & Hill, 2015)
(also see NY Times article: *Black, Deaf, and Extremely Online*, Jan 31, 2021)

Let's think about
variation in early life experience

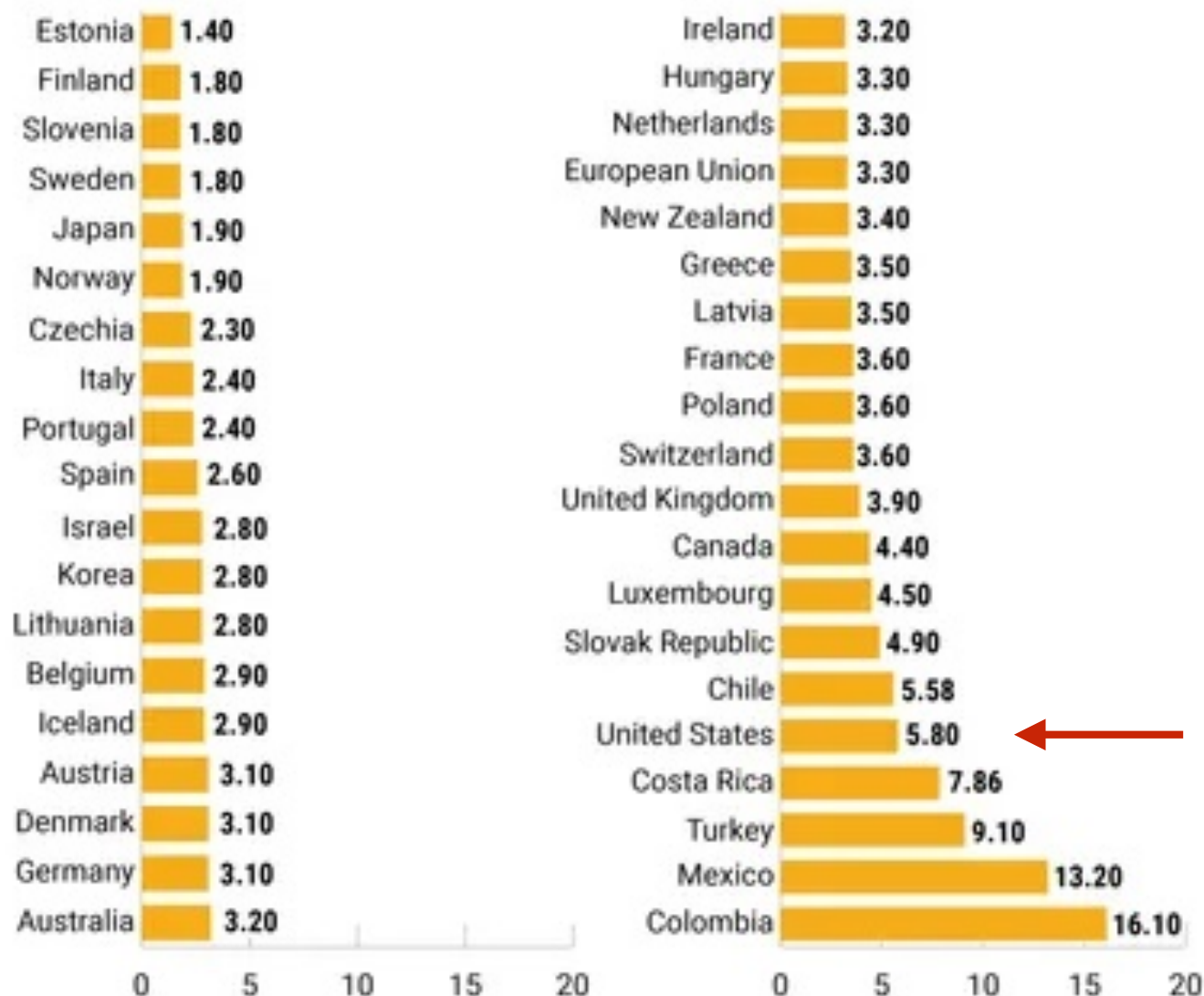
Social class: one's access to social and economic resources as well as one's rank relative to others in society.

Class is not just something impersonal, which you are born into; it is something that is done to you until it takes hold.

- Jedediah Purdy

Infant mortality in the OECD

Number of infant deaths for every 1,000 live births.



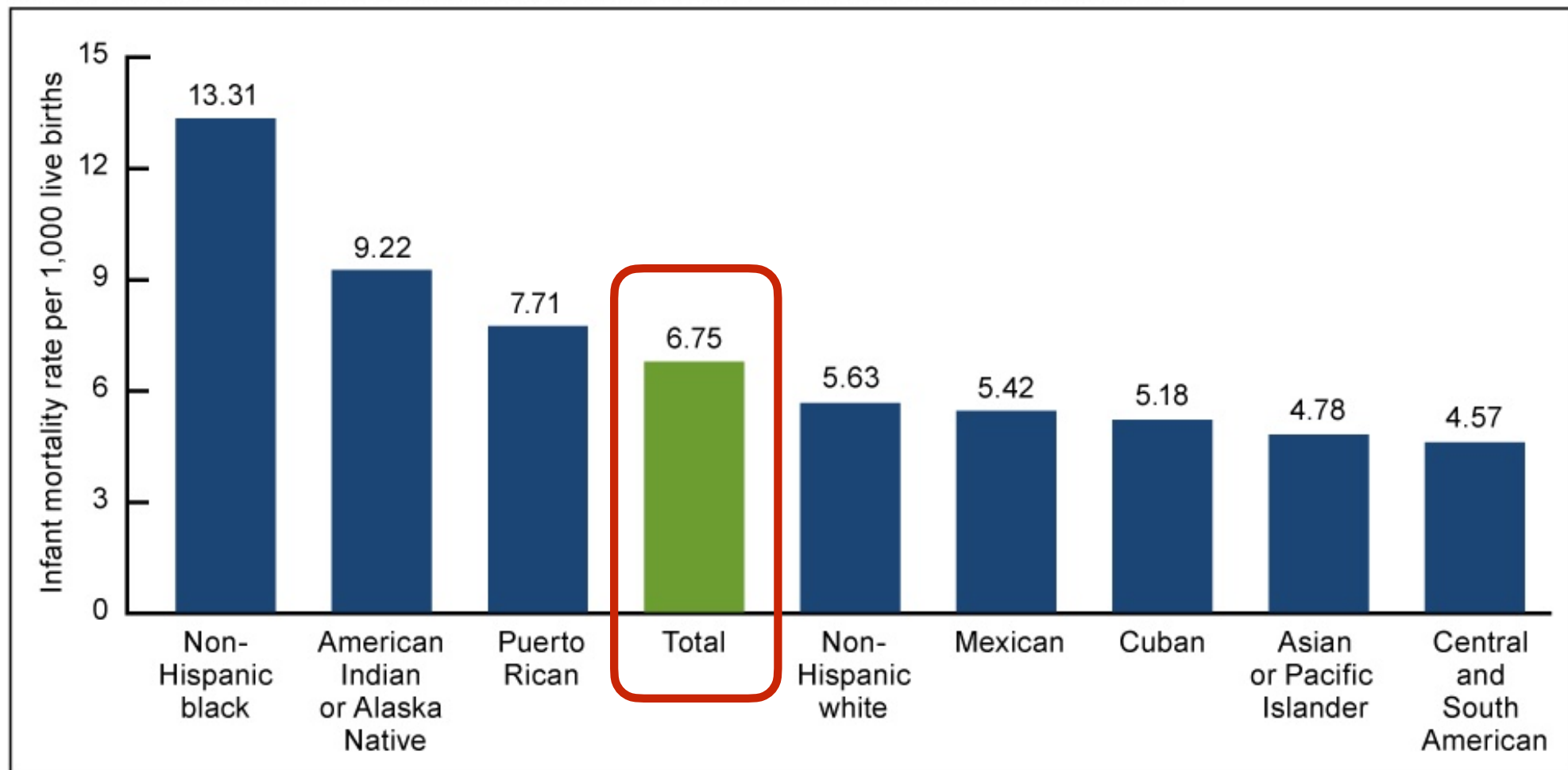
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Note: Most recent data available for each country, 2015-2022

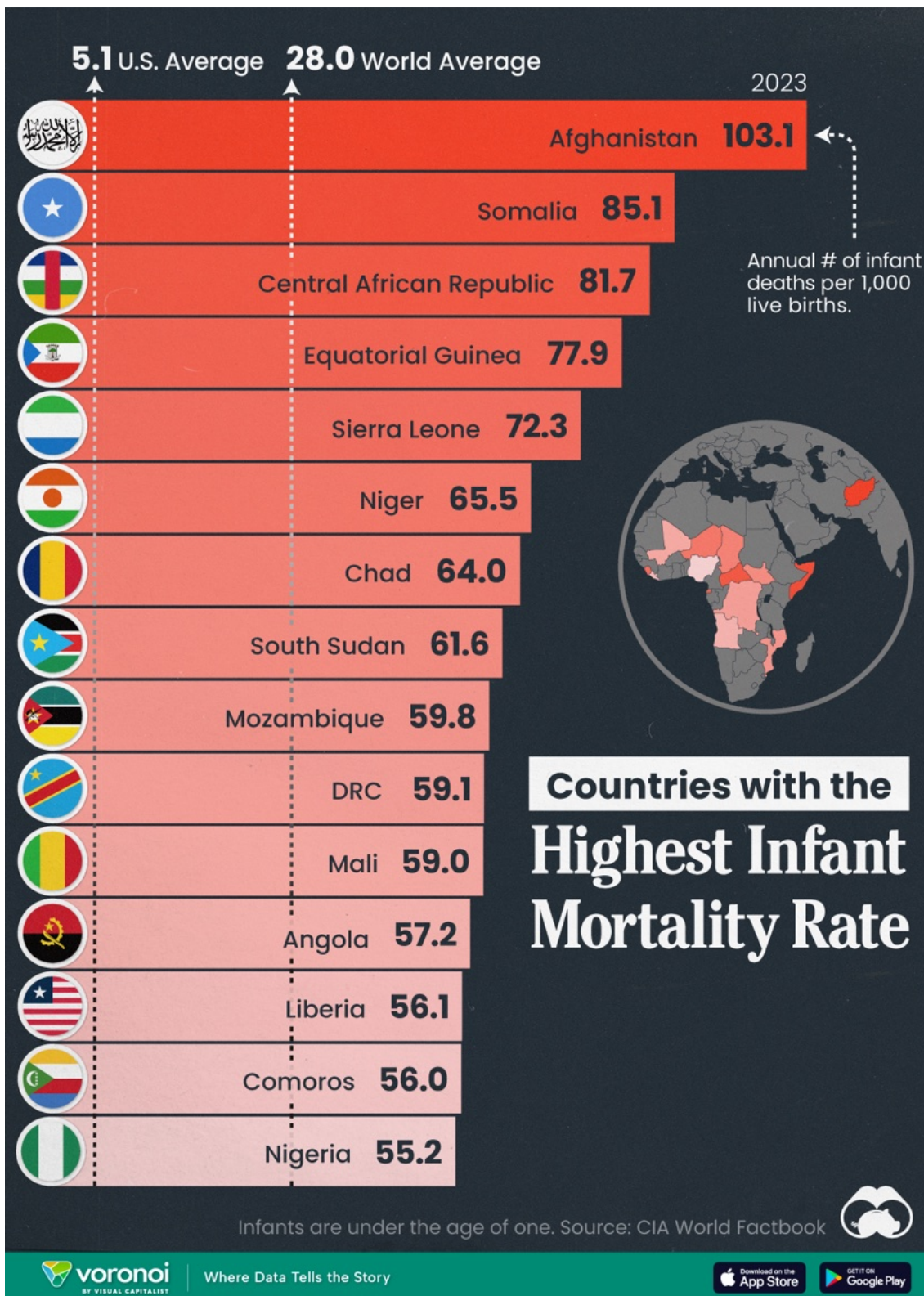
Source: OECD

GZERO
everything is political

Figure 1. Infant mortality rates, by race and Hispanic origin of mother: United States, 2007



SOURCE: CDC/NCHS, linked birth/infant death data set, 2007.





Praesis et Coratores Vniversitates Princetoninsis

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Paul M. Dell
Praesis

Henry H. Ruth
Scribo







U.S. child poverty in 1964: 23%

U.S. child poverty in 2016: 21%

U.S. child poverty in 2021: 5%

U.S. child poverty in 2024: 16%

March 9, 2021

A Safety Net for American Children

Inside the \$1.9 trillion economic stimulus bill, there is a provision that is revolutionary in the way the U.S. government regards minors.

Gap Between Rich And Poor Named 8th Wonder Of The World

1/24/11 8:00am • SEE MORE: WORLD ▾



The awe-inspiring gap.

PARIS—At a press conference Tuesday, the World Heritage Committee officially recognized the Gap Between Rich and Poor as the "Eighth Wonder of the World," describing the global wealth divide as the "most colossal and enduring of mankind's creations."

"Of all the epic structures the human race has devised, none is more staggering or imposing than the Gap Between Rich and Poor," committee chairman Henri Jean-Baptiste said. "It is a tremendous, millennia-old expanse that fills us with both wonder and humility."

Rising Income Inequality Causing Wealthy Americans To Take On Second Sailboat

NEWS IN BRIEF • Rich • Money • News • ISSUE 50•49 • Dec 11, 2014



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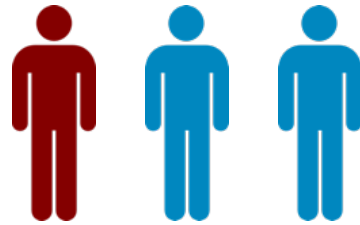
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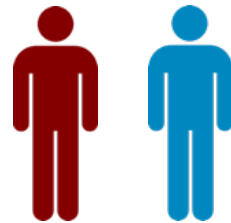


PALO ALTO, CA—Explaining how economic conditions were drastically altering citizens' behavior and spending patterns, a report released Wednesday by researchers at Stanford Business School revealed that rising income inequality was causing more wealthy Americans to take on a second sailboat. "While those with higher incomes used to be able to get by with just a

Kids in the U.S. today...



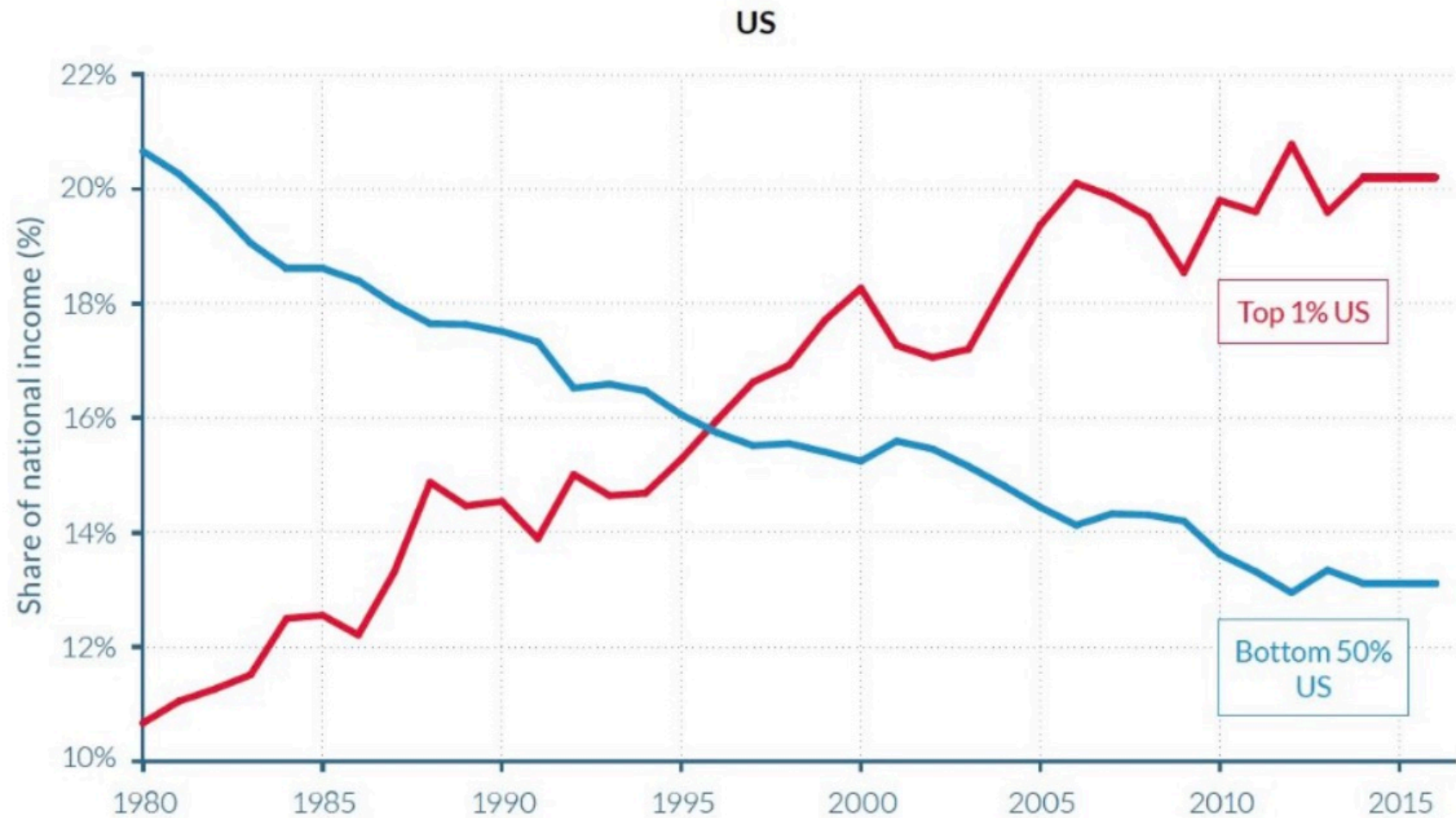
1 in 3 are forced to rely on food stamps



Nearly 1 out of 2 children live in families near the poverty line

...plus, income inequality has gotten dramatically worse since the 1970s

Top 1% vs. Bottom 50% national income shares in the US and Western Europe, 1980–2016: Diverging income inequality trajectories



Source: WID.world (2017). See [wir2018.wid.world](#) for data series and notes.

In 2016, 12% of national income was received by the top 1% in Western Europe, compared to 20% in the United States. In 1980, 10% of national income was received by the top 1% in Western Europe, compared to 11% in the United States.

700 million

people live in extreme poverty
(getting by on less than \$2.15/day)

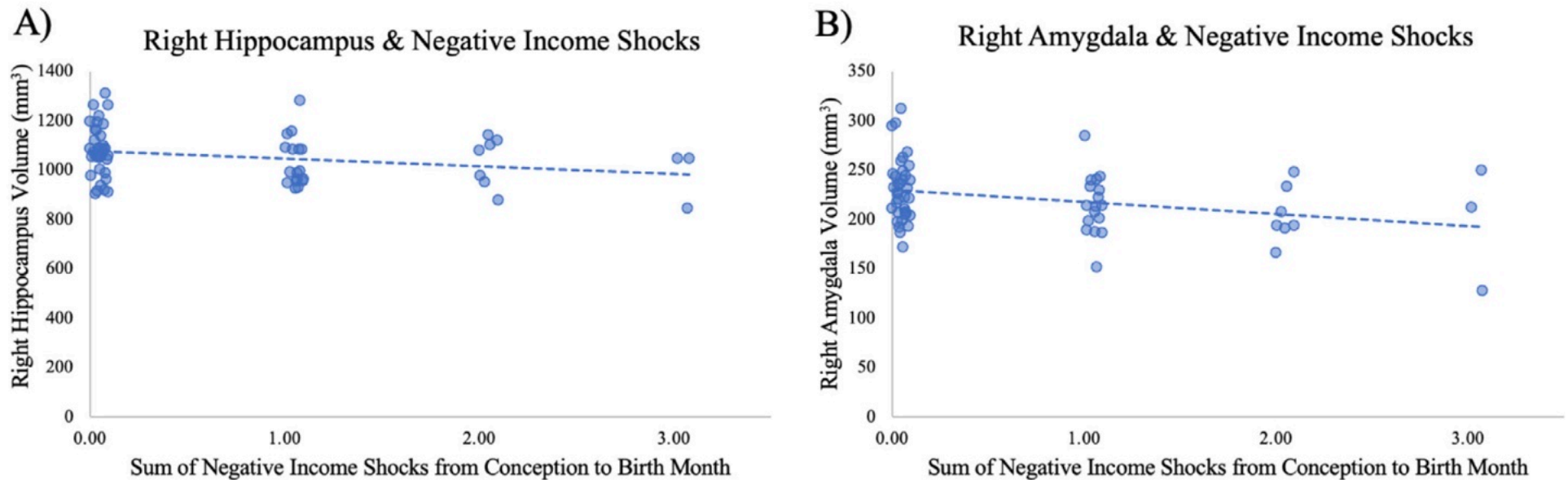
Minimum wage

A full-time minimum wage worker
can afford a one-bedroom rental
in only 7% of all U.S. counties

(i.e., 218 counties out of more than 3,000 nationwide)

Income instability predicts infant brain volume

Expectant parents who experience more financial shocks have newborns with smaller volume in right hippocampus and amygdala.
(controlling for birth weight)



Unpredictability (especially of monthly income) is known to be a core cause of stress.

Reducing income loss during the perinatal period is a promising solution.
(e.g., family and medical leave programs)

Environmental inequality



If you want a window into a context for childhood
that's probably different from your own,
find an hour to listen to this:

Harper High School

[http://www.thisamericanlife.org/radio-archives/episode/487/
harper-high-school-part-one](http://www.thisamericanlife.org/radio-archives/episode/487/harper-high-school-part-one)

Many domains of growth are fragile



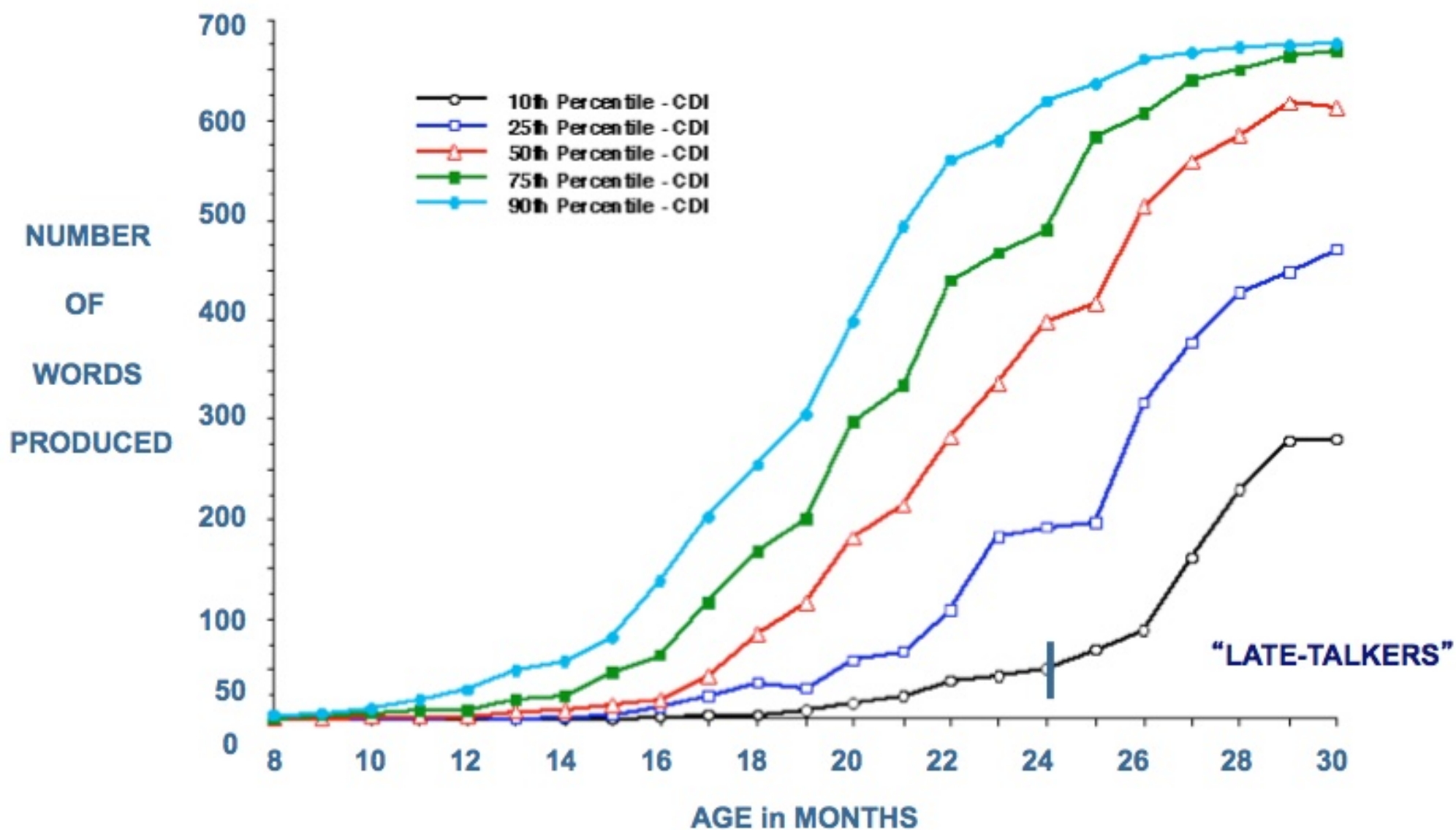
Poor fetal growth and stunting at age 2 predict worse outcomes

Early attachment relationships set the course for later social outcomes

Lack of language richness has detrimental, long-term effects on cognitive development



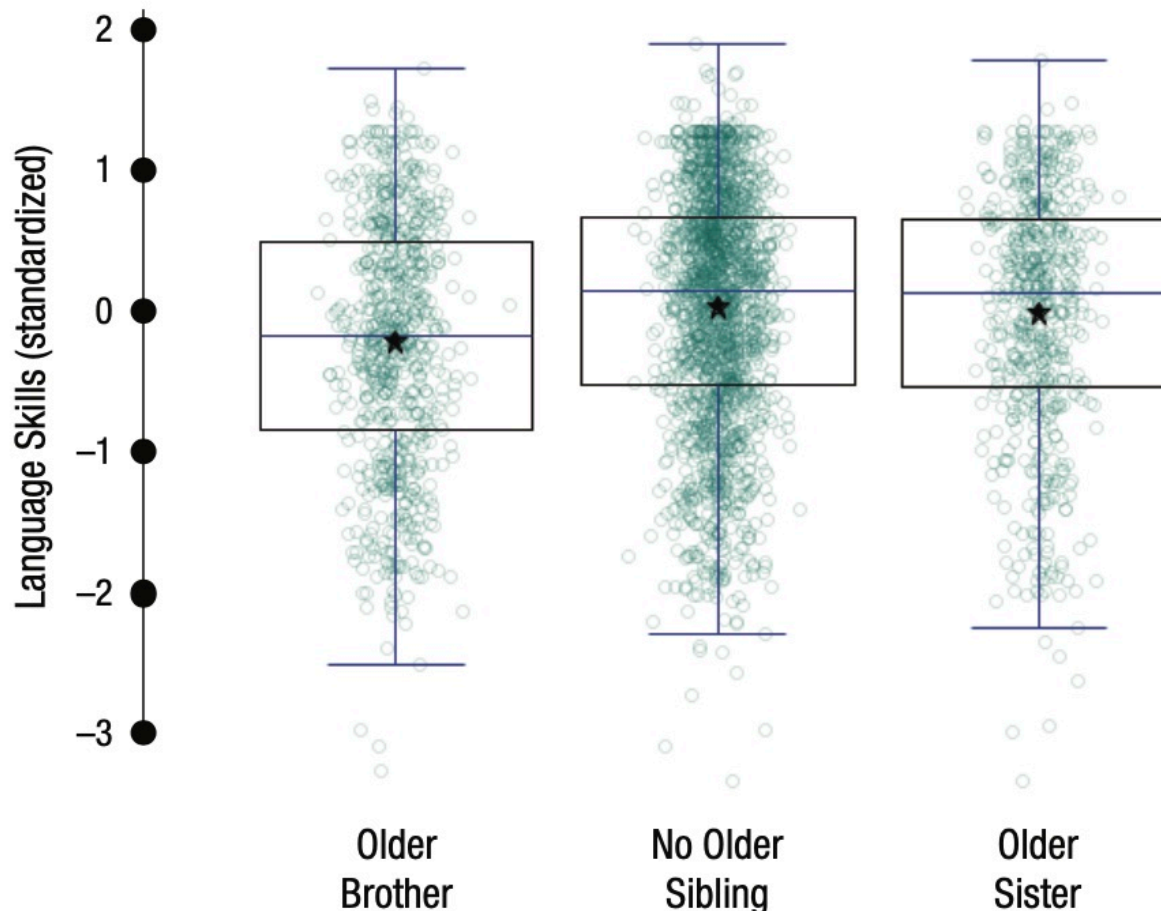
Individual differences: Language learning as a case study



Fenson et al. (2006)

Factors influencing individual differences

- **Cognitive differences:** working memory
- **Sex and birth order:** girls and first-borns have *slightly* larger vocabularies and use more complex language.

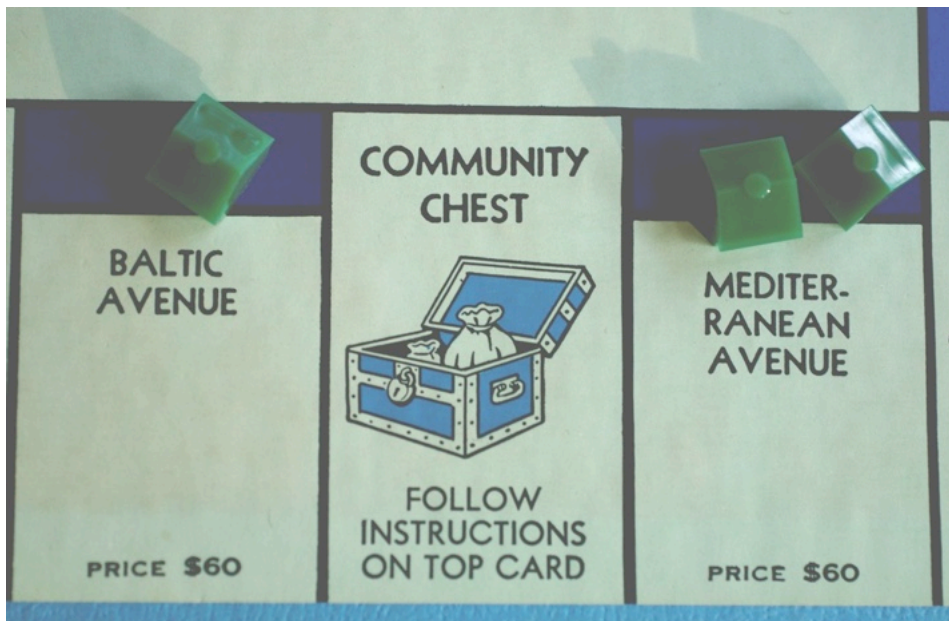


More older siblings = slightly lower vocab.



Factors influencing individual differences

- **Cognitive differences:** working memory
- **Sex and birth order:** girls and first-borns have larger vocabularies and use more complex language.
- **Socioeconomic status:** SES exerts a powerful influence on language outcomes. Why?



Factors influencing individual differences

- **Cognitive differences:** working memory
- **Sex and birth order:** girls and first-borns have larger vocabularies and use more complex language.
- **Socioeconomic status:** SES exerts a powerful influence on language outcomes. Why?
 - Nutrition, healthcare, enriching experiences...



How many words do kids actually hear per year?

Probably ~200 utterances per hour on average.

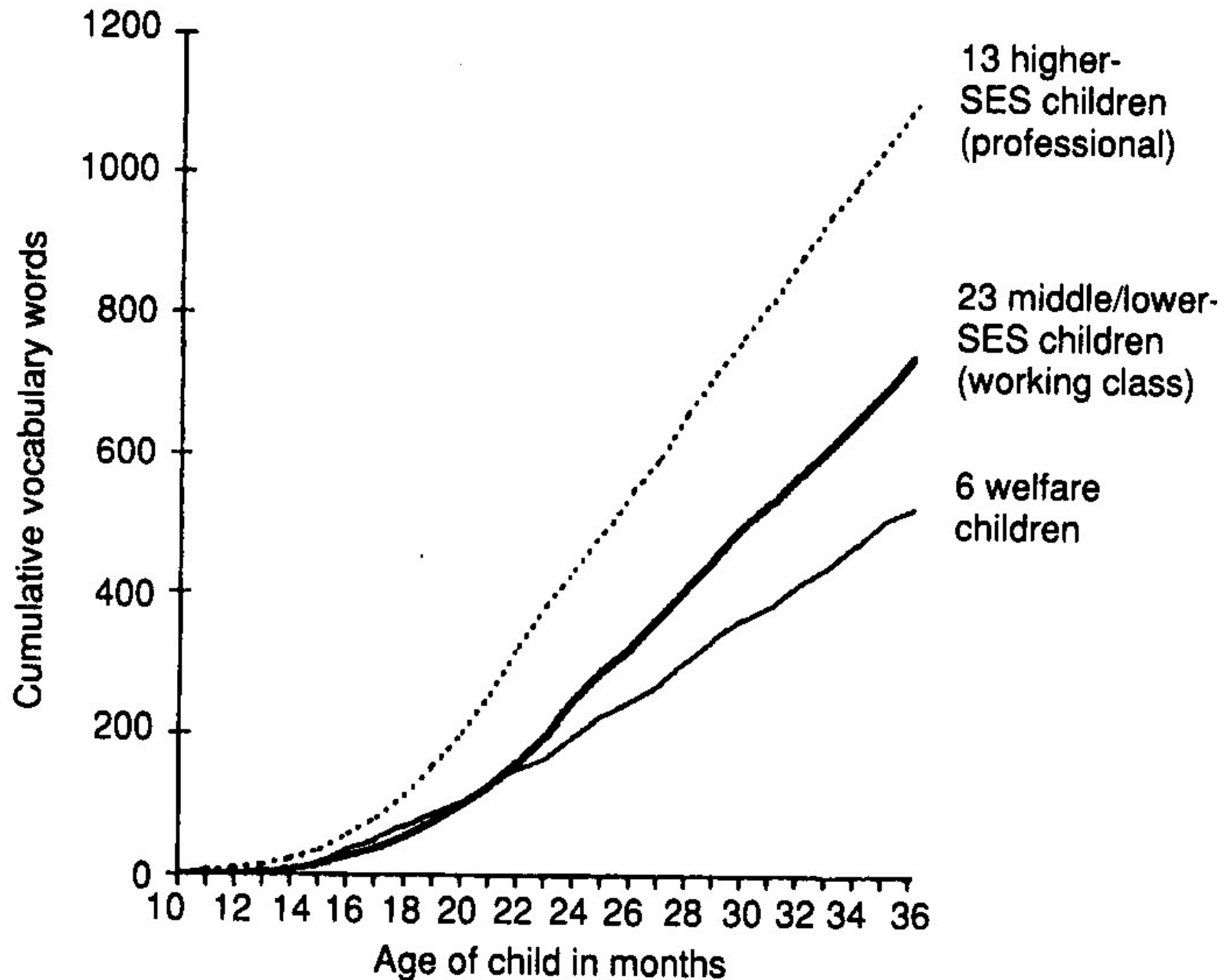
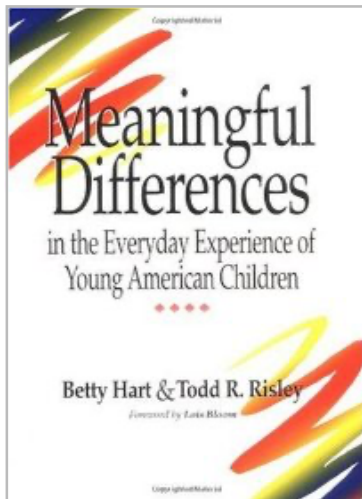
200 utterances/hour

- * 10.5 hours awake/day
- * 365 days/year
- * 5 words per utterance

= ~4 million words per year

= ~16 million words by the 4th birthday

Hart & Risley's (1995) Kansas City sample



Does household language experience vary across by SES?

By 4 years, children from high-SES families may hear, on average, slightly more words and higher-quality speech than children from low-SES families.

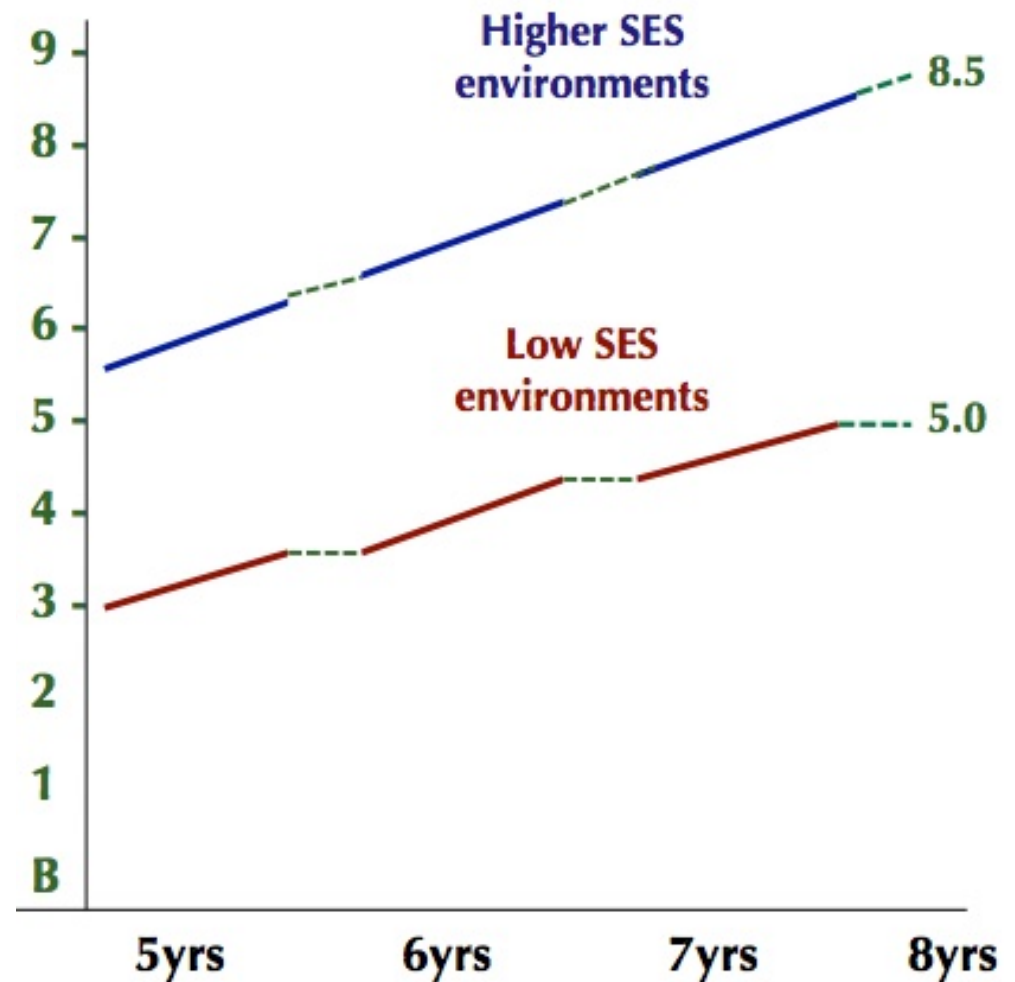
- Estimates of this difference range from 1% to 35%
(Gilkerson et al., 2017; Hart & Risley, 1995; Hoff, 2006; Sperry et al., 2019)
- No difference when you examine a global sample.
(Bergelson et al., 2023)
- Variation is huge within a given socioeconomic tier

The jury is still out on whether there are SES-related differences or not, and whether any differences actually matter.

What do you think? Are we barking up the wrong tree when we focus on language experiences of higher- vs. lower-SES kids? What experiential differences should we focus on?

Environmental differences predict later academic outcomes and language abilities

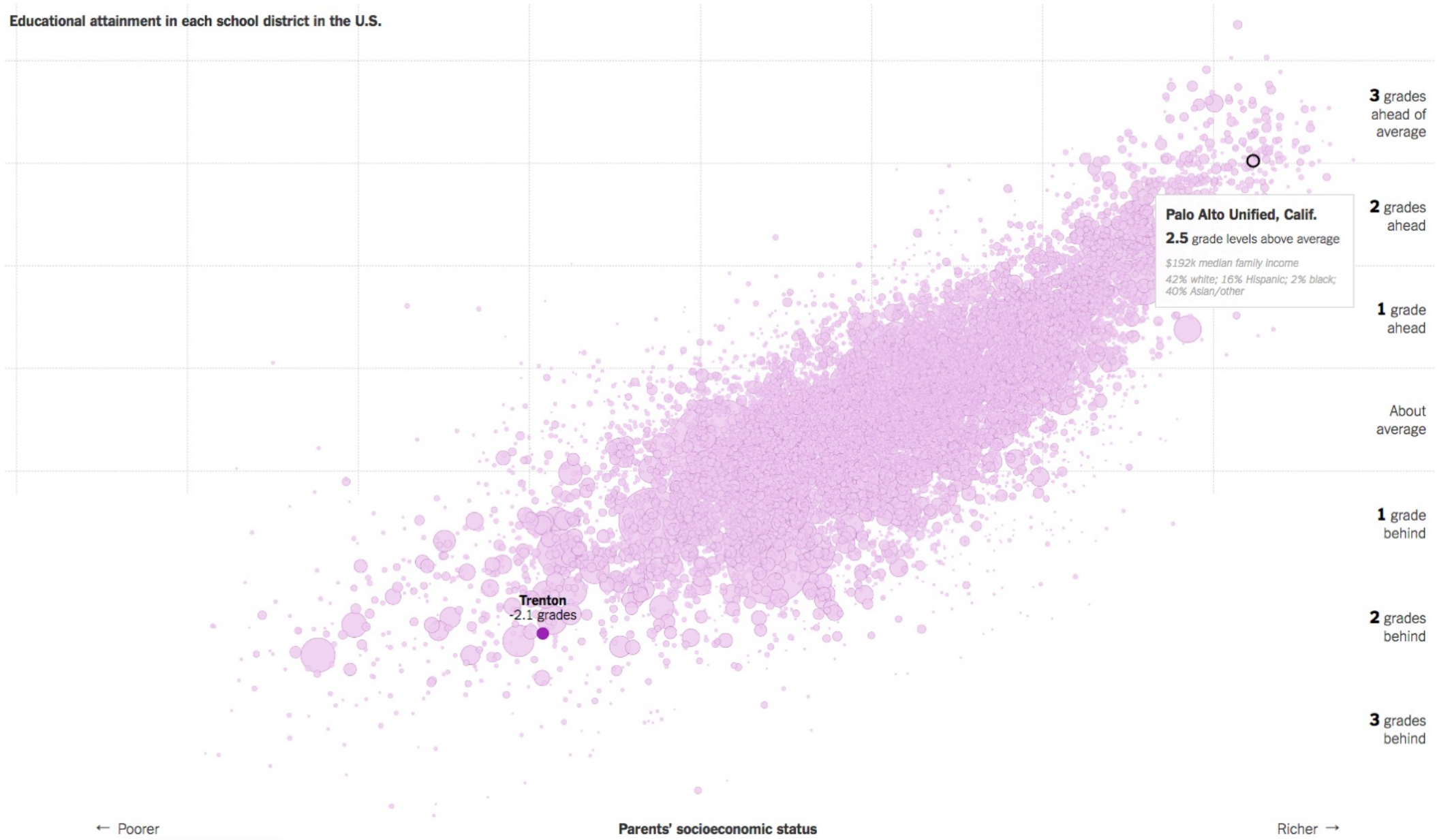
Developmental Age
(based on measures of
cognitive and language Skills)



Sixth graders in the richest school districts are four grade levels ahead of children in the poorest districts. [RELATED ARTICLE](#)

Trenton, N.J.

Educational attainment in each school district in the U.S.



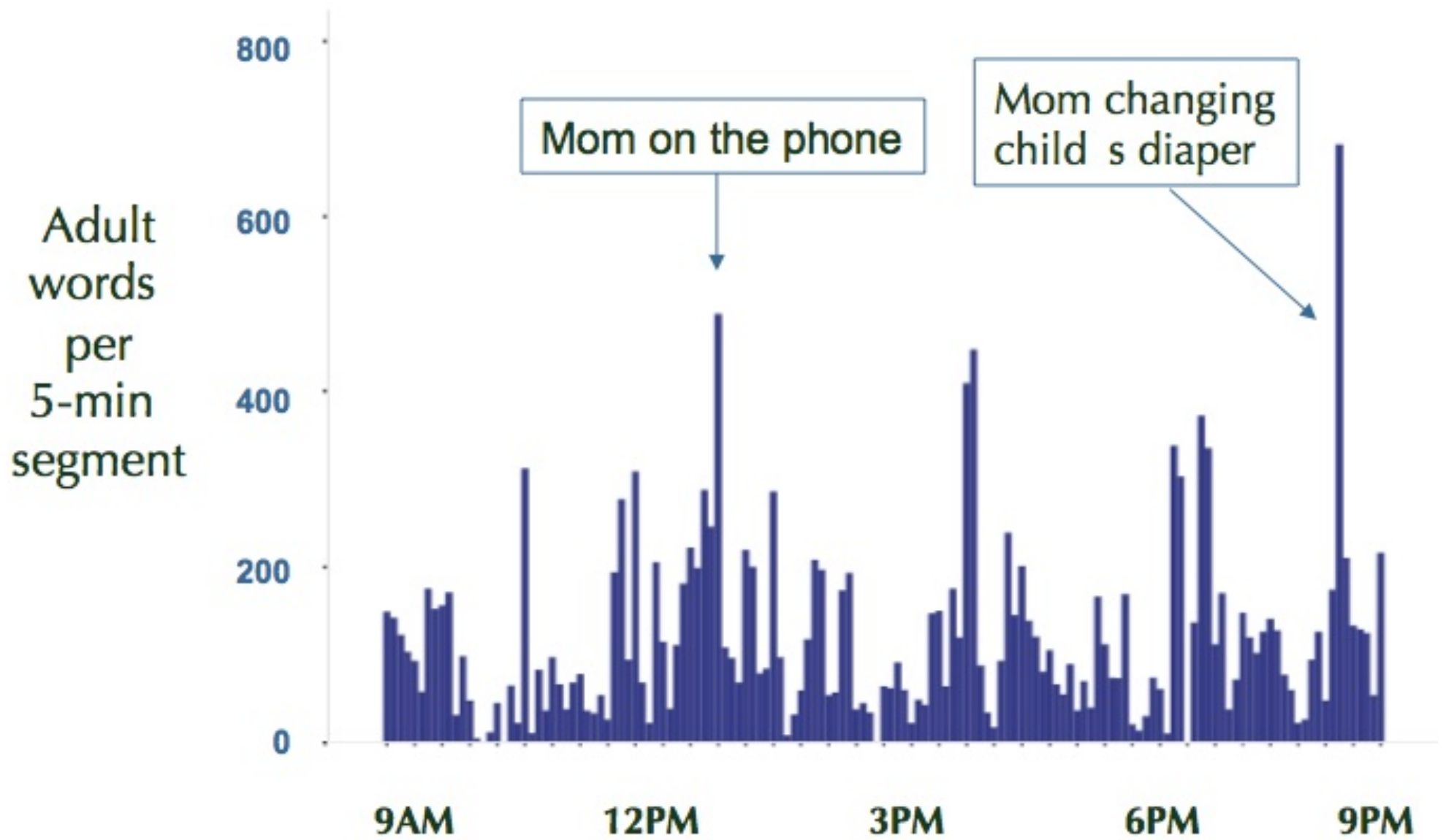
Bridge Apartment Study, NYC, 1973

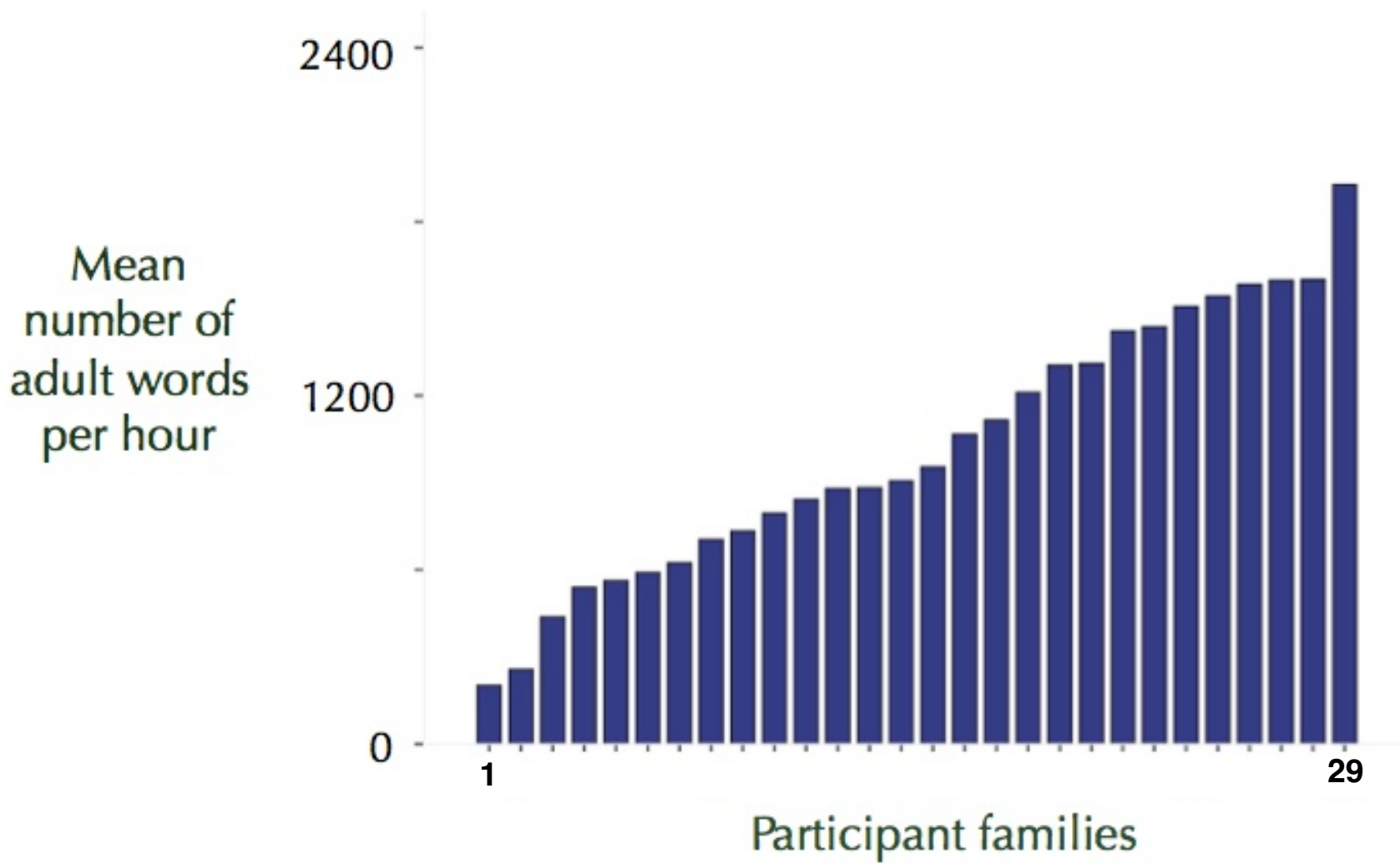


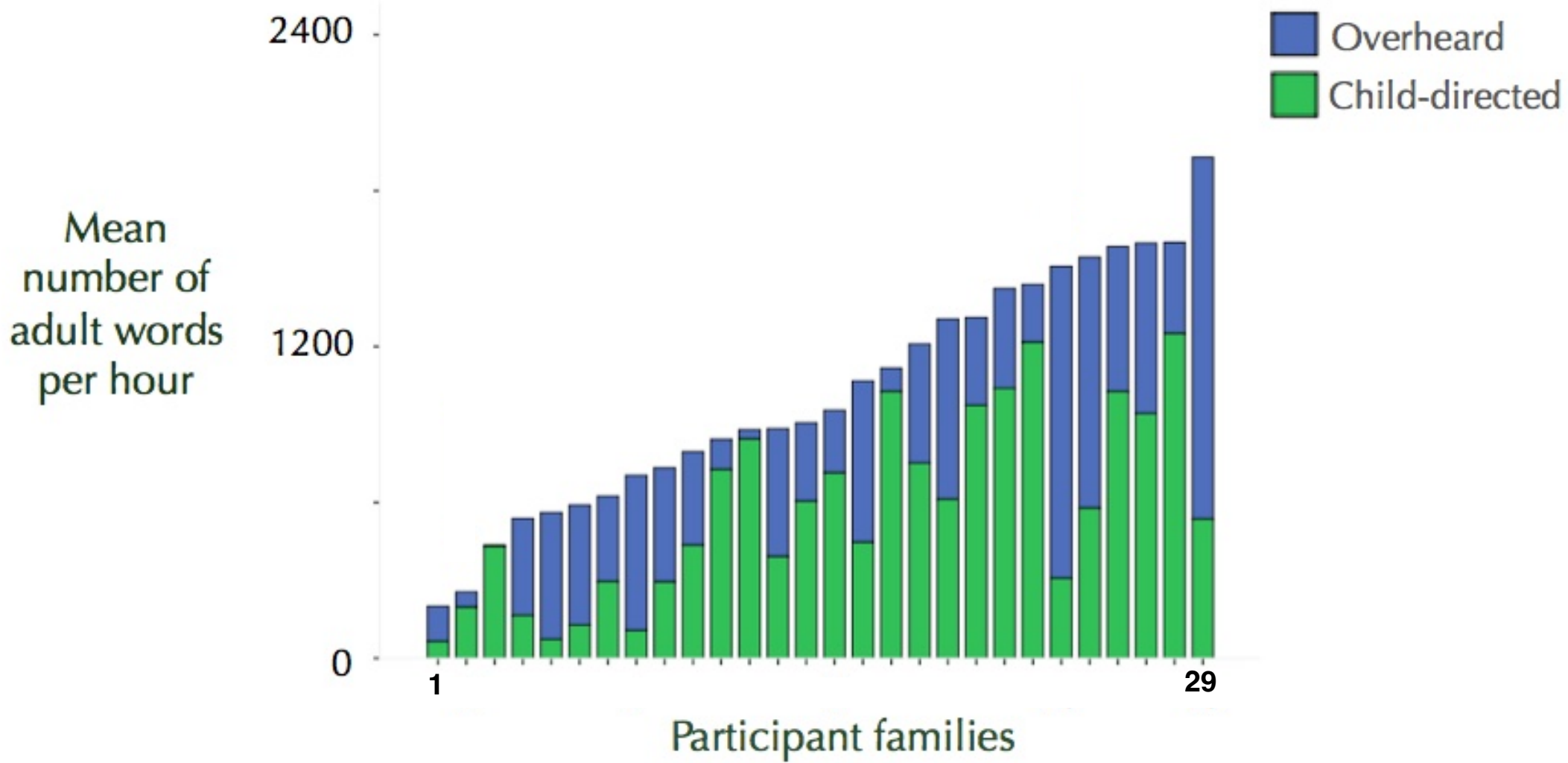
(see also, Ha, Yeung, et al., 2019, *Environmental Research*)

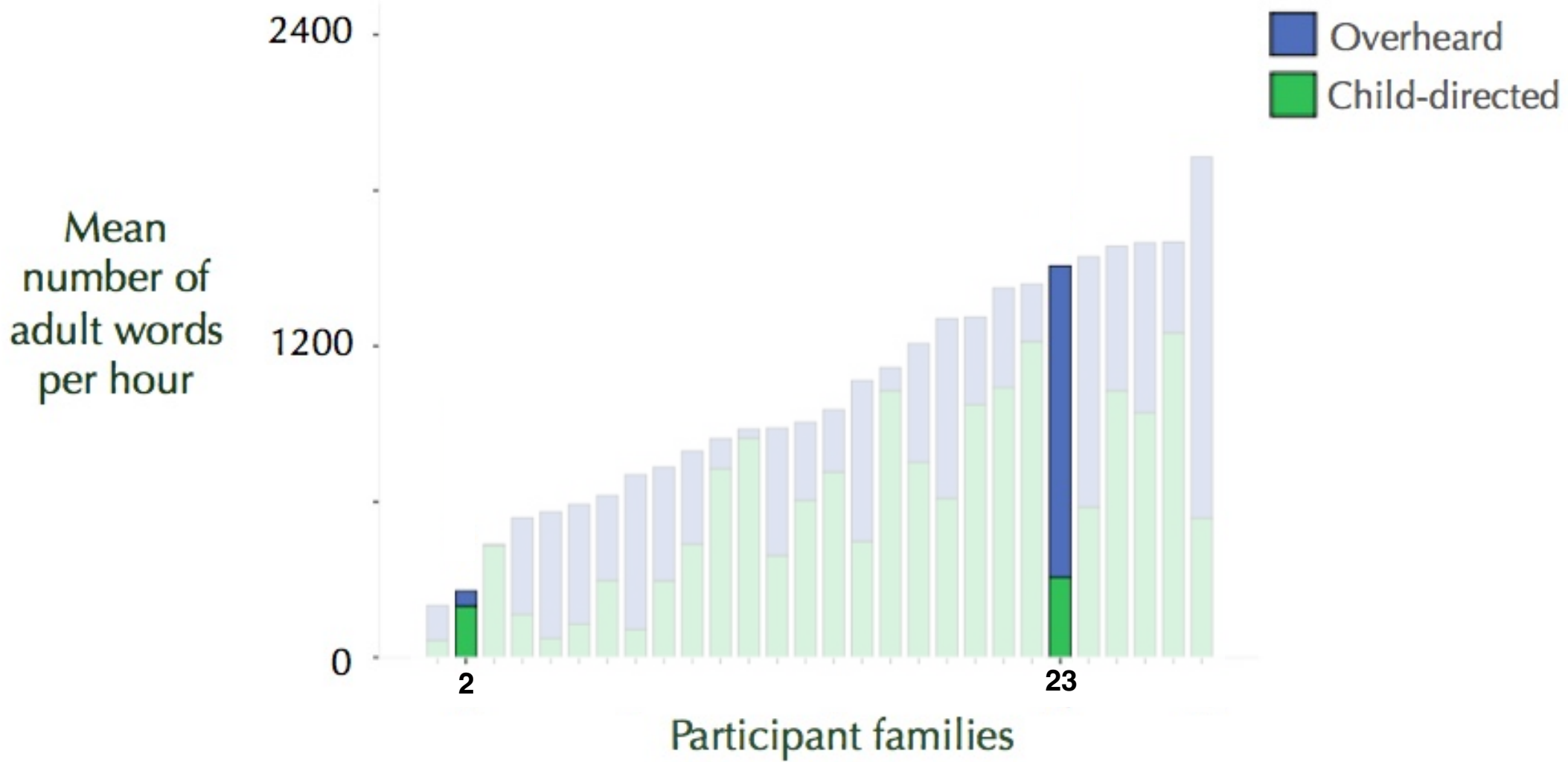


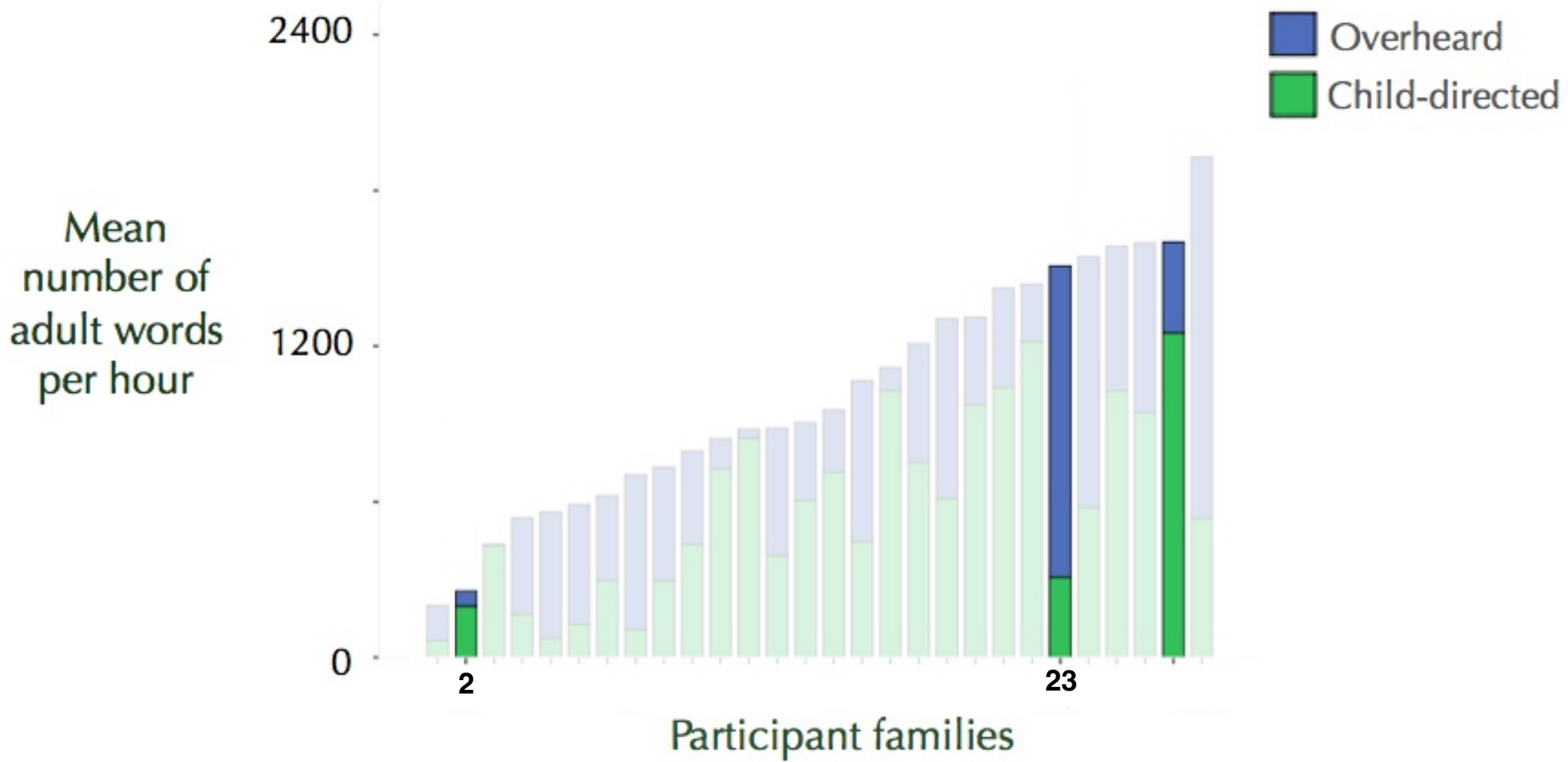
Study on language variability (Weisleder et al., 2013)
LENA recorders captured home language use
Automated counts of words and conversational turns
29 low-income Latine families











Child-directed speech drives gains in real-time language processing that promote further vocabulary growth



This provides a possible **behavioral mechanism** underlying SES disparities in children's language skills.

Children who experience more **conversational turns** with adults exhibit greater left inferior frontal (Broca's area) activation.

(controlling for SES, IQ, and overall talkativeness)

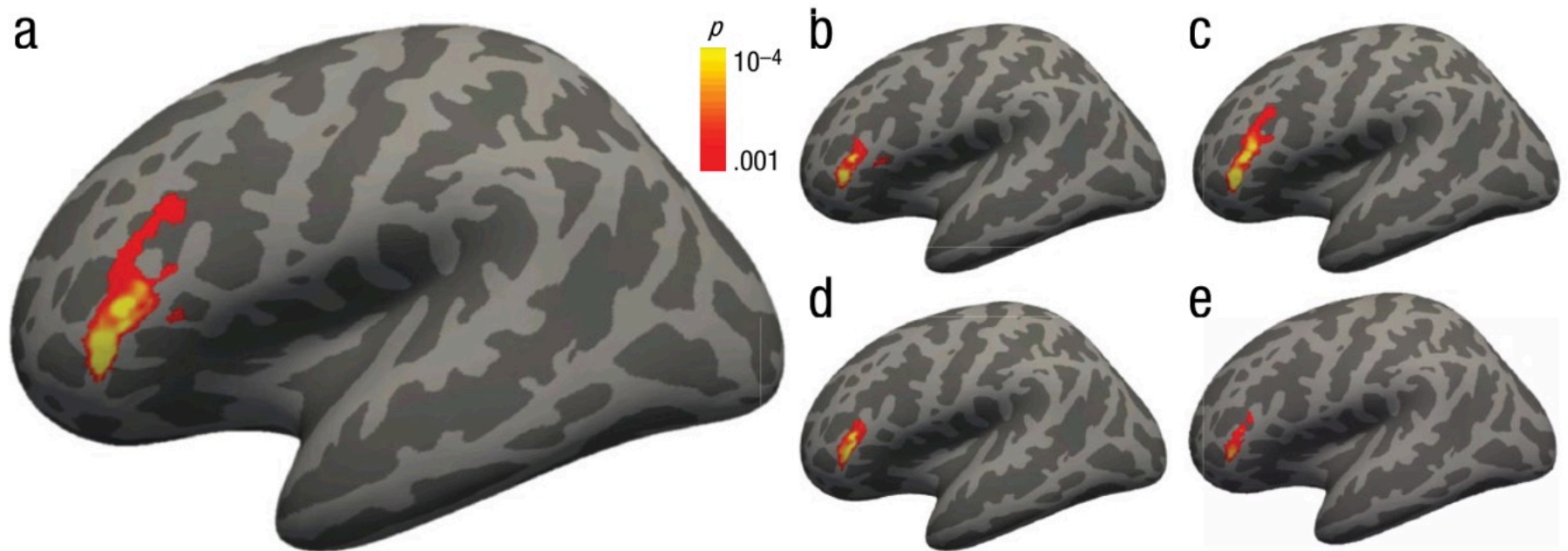
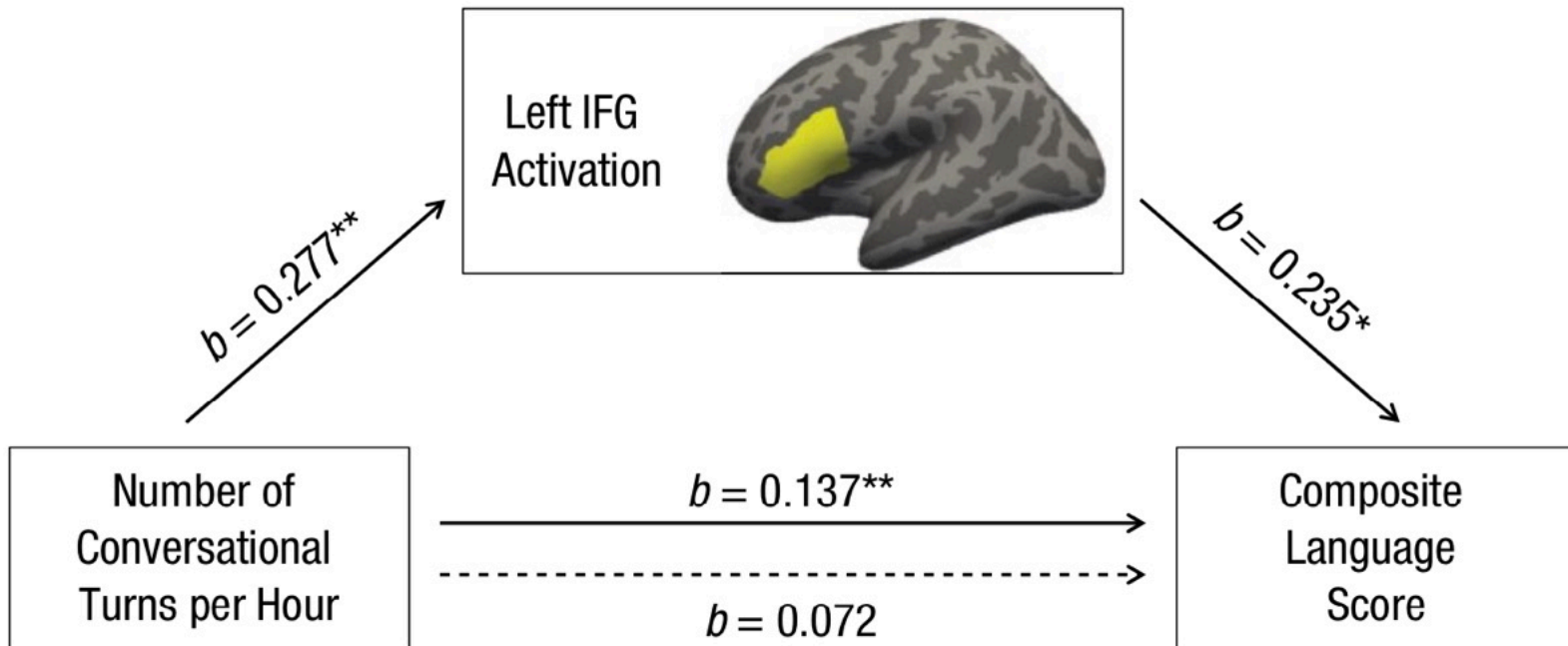


Fig. 5. Correlations between activation during language processing and the number of hourly conversational turns children experienced. The brain image in (a) shows the zero-order correlation between the number of conversational turns and activation in the forward > backward speech contrast. Correlations remained significant when controlling for (b) parental education and income, (c) verbal and nonverbal assessment scores, (d) individual numbers of adult words and child utterances, and (e) all of these covariates.

Turn-taking during natural interactions promotes sustained attention, working memory, and language

Children who experience more **conversational turns** with adults exhibit greater left inferior frontal (Broca's area) activation.



This provides a possible **neural mechanism** underlying variation in children's language skills.

This research has made its way into the media
(SVU, Orange is the New Black, Telemundo, etc.)



Experience shapes outcomes: The case of google-era kids

<https://www.youtube.com/watch?v=zhPklt9nYas>



Experience shapes outcomes: The case of Romanian orphans

http://www.youtube.com/watch?v=bvL_DGjGuhA





Providence's \$5 million plan to shrink the "word gap"

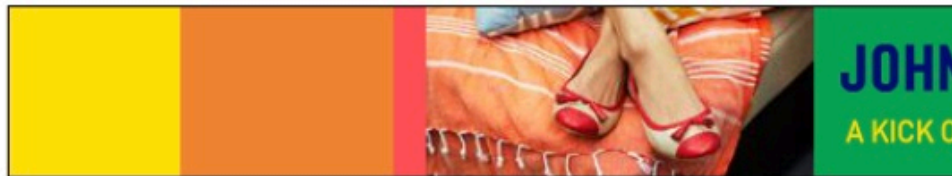
Can a city boost achievement by recording, and changing, how parents talk to kids?

By Ben Zimmer | GLOBE CORRESPONDENT MARCH 24, 2013

The New York Times

The Opinion Pages

WORLD U.S. N.Y. / REGION BUSINESS TECHNOLOGY SCIENCE HEALTH SPORTS OPINION



Opinionator

Exclusive Online Commentary From The Times

FIXES | April 10, 2013, 3:25 pm | 499 Comments

The Power of Talking to Your Baby

By TINA ROSENBERG



Fixes looks at solutions to social problems and why they work.

By the time a poor child is 1 year old, she has most likely already fallen behind middle-class children in her ability to talk, understand and learn. The gap between poor children and wealthier ones widens each year, and by high school it has become a chasm. American attempts to close this gap in schools have largely failed, and a

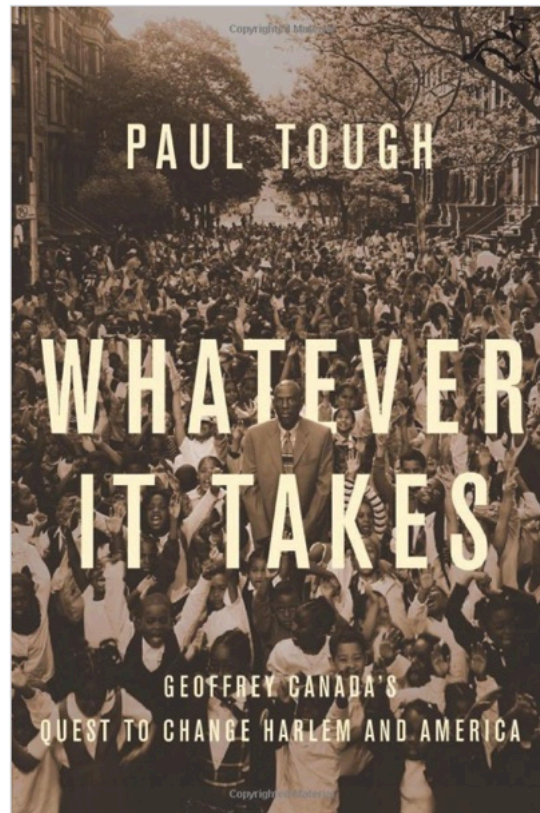


Harlem Children's
ZONE



A national model
for breaking the
cycle of poverty

EDUCATION + FAMILY & COMMUNITY + HEALTH = RESULTS



To Help Language Skills of Children, a Study Finds, Text Their Parents With Tips

By MOTOKO RICH NOV. 14, 2014



A prekindergarten class in San Francisco. A study found preschoolers benefited when their parents received texts with guidance on how to help them. Jason Henry for The New York Times



Email



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With research showing [language gaps between the children of affluent parents and those from low-income families](#) emerging at an early age, educators have puzzled over how best to reach parents and guide them to do things like read to their children and talk to them regularly.

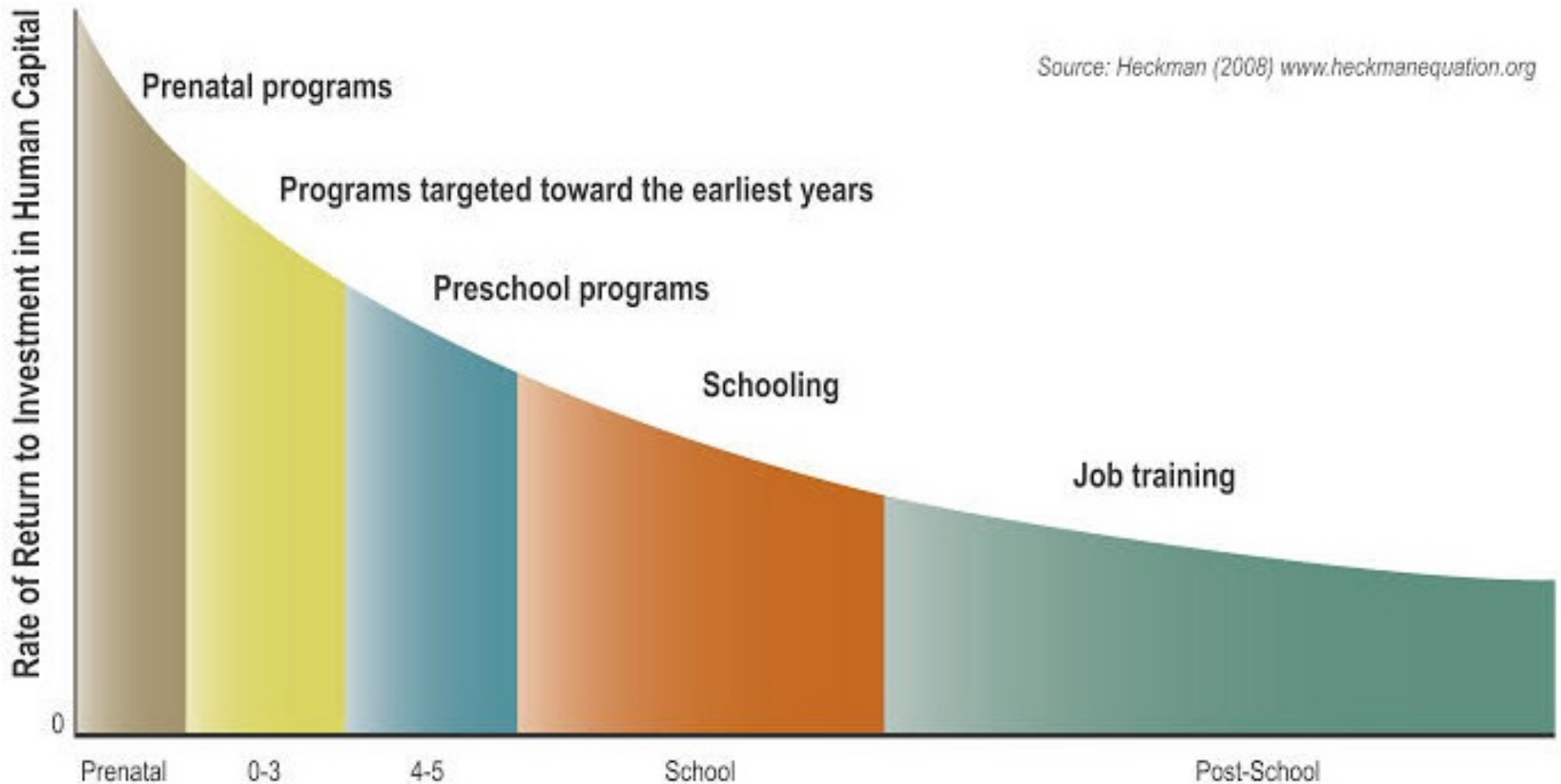
A new [study](#) shows that mobile technology may offer a cheap and effective solution. The research, released by the [National Bureau of Economic](#)

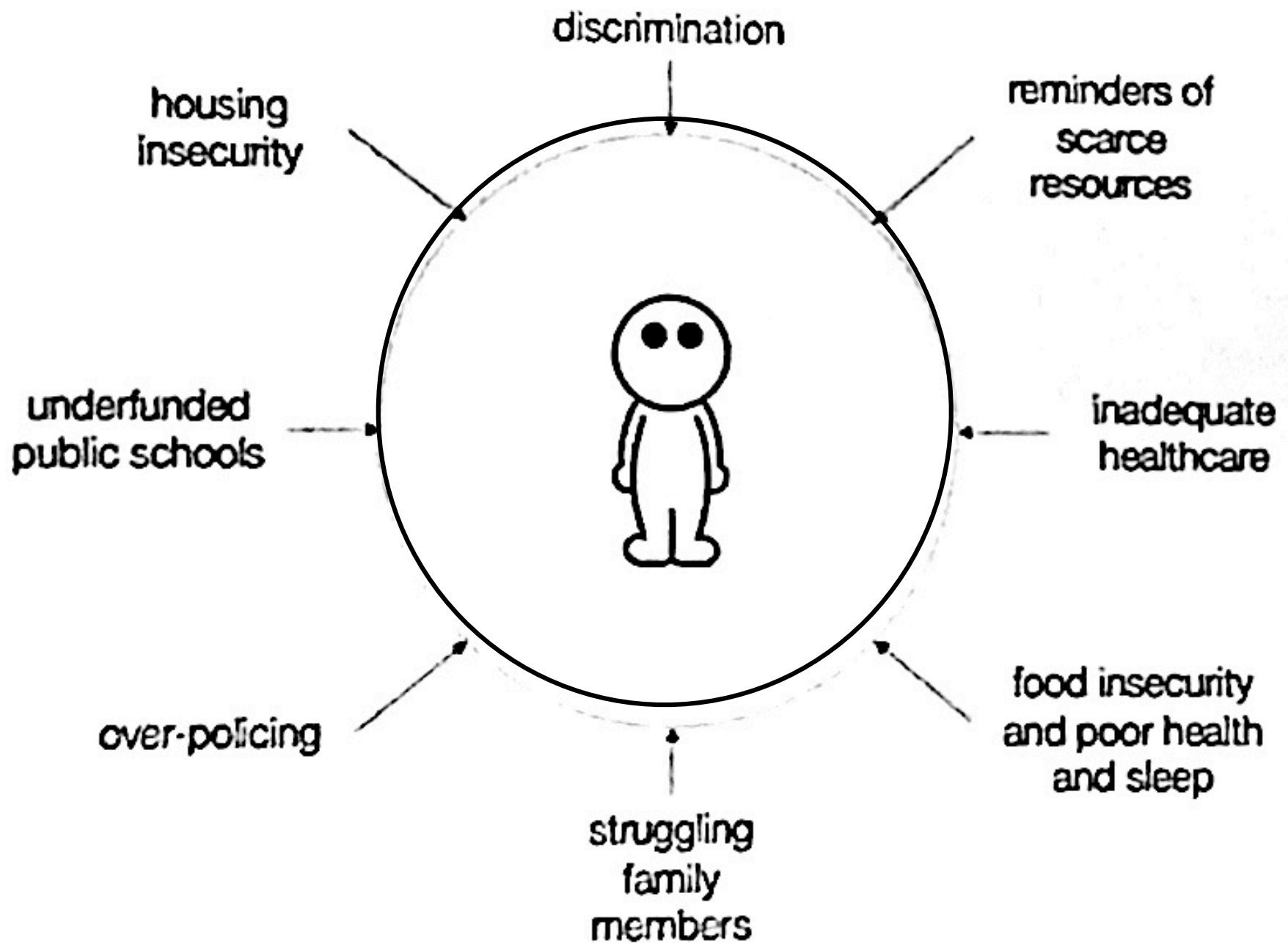
At least in W.E.I.R.D. cultures, infants benefit from rich, varied, contingent, and playful interactions.

Increased opportunities for fun communication can alter the course of development

...with cascading advantages for children.

“So if politicians are genuinely looking for a bipartisan issue to break through the Washington gridlock, here’s a suggestion: invest in early education”





Forging a Pediatric Primary Care–Community Partnership to Support Food-Insecure Families

Q:1,2

AUTHORS: Andrew F. Beck, MD, MPH,^{a,b} Adrienne W. Henize, JD,^a Robert S. Kahn, MD, MPH,^a Kurt L. Reiber, JD, MBA,^c John J. Young, MEd,^c and Melissa D. Klein, MD, MEd^{a,b}

Divisions of ^aGeneral and Community Pediatrics and ^bHospital Medicine, Department of Pediatrics, Cincinnati Children's Hospital Medical Center, Cincinnati, Ohio; and ^cFreestore Foodbank, Cincinnati, Ohio

KEY WORDS

food insecurity, hunger, pediatric, primary care

abstract

BACKGROUND AND OBJECTIVES: Academic primary care clinics often care for children from underserved populations affected by food insecurity. Clinical-community collaborations could help mitigate such risk. We sought to design, implement, refine, and evaluate Keeping Infants Nourished and Developing (KIND), a collaborative intervention focused on food-insecure families with infants.

Learning and development
are fragile, and subject to the
environment

This is a humanitarian
and pragmatic problem.

Developmental science can help
to target and refine effective interventions
for disadvantaged youth.